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ITA0609

Exp-20: Implement Future Sales Prediction using a suitable machine learning algorithm

```
# Import required libraries

from sklearn.datasets import fetch_california_housing

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

# Load sample dataset

data = fetch_california_housing()

X = data.data

y = data.target

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create the Linear Regression model

model = LinearRegression()

# Train the model

model.fit(X_train, y_train)

# Predict future values

y_pred = model.predict(X_test)

# Display results

print("Actual Sales:")

print(y_test[:10])

print("\nPredicted Sales:")

print(y_pred[:10])

# Evaluate the model

print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
```

```python
print("R2 Score:", r2_score(y_test, y_pred))
```

Output:

```
Actual Sales:
[0.477   0.458   5.00001 2.186   2.78    1.587   1.982   1.575   3.4
 4.466  ]

Predicted Sales:
[0.71912284 1.76401657 2.70965883 2.83892593 2.60465725 2.01175367
 2.64550005 2.16875532 2.74074644 3.91561473]

Mean Squared Error: 0.5558915986952422
R2 Score: 0.5757877060324524
```